

NOR Gate

Truth table

NOR Gate		
a	b	y
0	0	1
0	1	0
1	0	0
1	1	0

Behavioral source code (nor_gate.v)

```
module nor_gate(y,a,b);
output y;
input a,b;
assign y=~(a|b);
endmodule
```

Testbench source code (nor_gate_tb.v)

```
`timescale 1ns/1ps

module nor_gate_tb;
    // 1. Creating registers for inputs and a wire for the output
    reg test_a;
    reg test_b;
    wire test_y;

    // 2. Instantiating the actual NOR gate module (connect pins)
    nor_gate uut (
        .y(test_y),
        .a(test_a),
        .b(test_b)
    );
```

```
// 3. Generating the input signals over time
initial begin
    // Telling iverilog to dump waveform data for GTKWave
    $dumpfile("simulation.vcd");
    $dumpvars(0, nor_gate_tb);

    // Test Case 1: Both 0;
    test_a = 0; test_b = 0; #5;

    // Test Case 2: 0 and 1;
    test_a = 0; test_b = 1; #10;

    // Test Case 3: 1 and 0;
    test_a = 1; test_b = 0; #15;

    // Test Case 4: 1 and 1;
    test_a = 1; test_b = 1; #1.5;

    $finish; // End simulation
end
endmodule
```

```
call C:\oss-cad-suite\environment.bat
```

fgfg

```
iverilog -o simulation.vvp nor_gate.v nor_gate_tb.v
```

fgfg

```
vvp simulation.vvp
```

Generate & check output waveform - Fig. 4(a)

```
gtkwave simulation.vcd
```

```
yosys
```

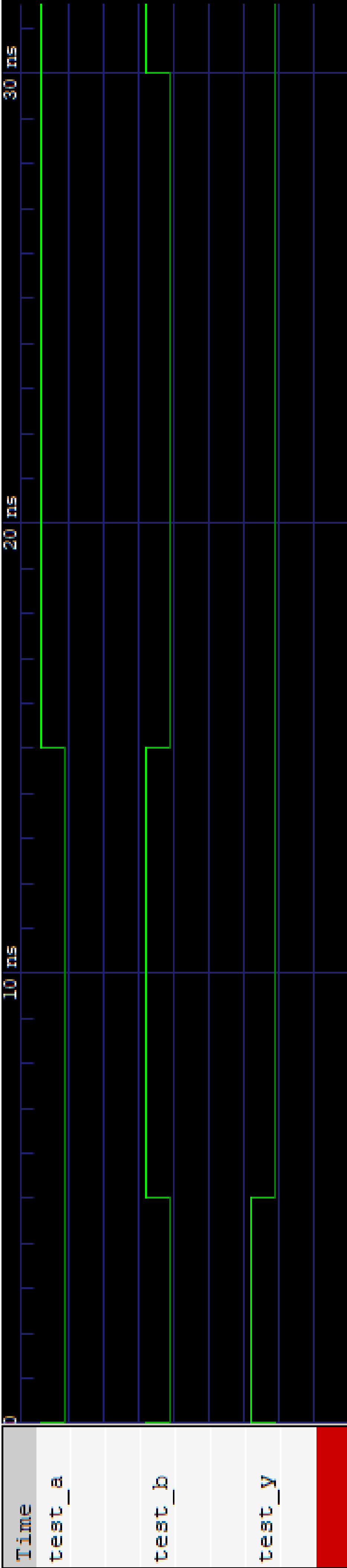


Fig. 4(a) - NOR gate output waveform

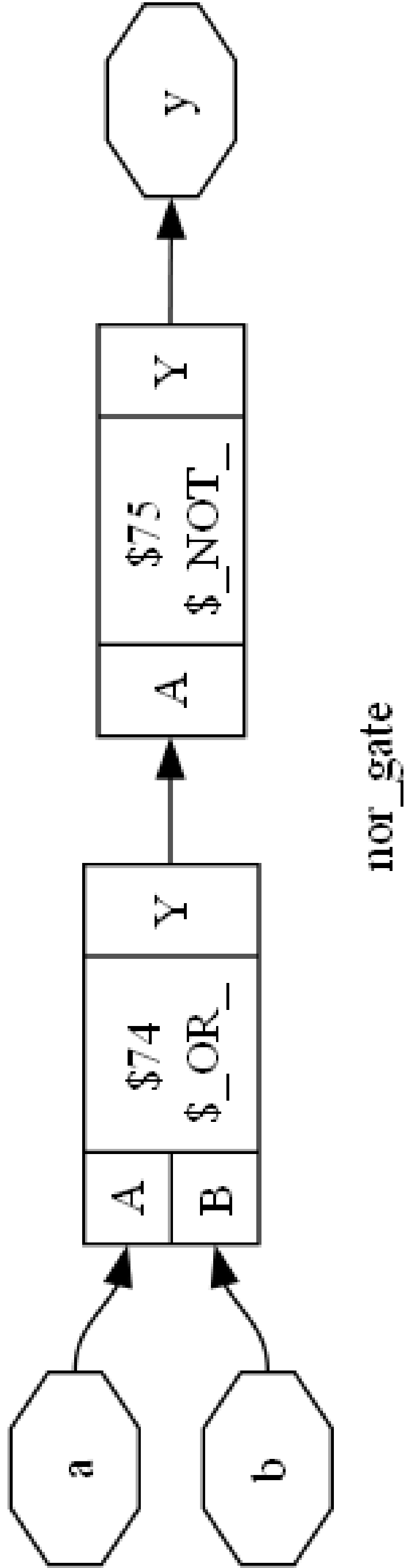


Fig. 4(b) - NOR gate generated schematic

```
/-----\
| yosys -- Yosys Open SYnthesis Suite                               |
| Copyright (C) 2012 - 2026  Claire Xenia Wolf <claire@yosyshq.com> |
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\-----/
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
```

```
yosys> read_verilog nor_gate.v
```

```
yosys> hierarchy -top nor_gate
```

```
yosys> proc opt techmap opt
```

```
yosys> write_verilog nor_gate_netlist.v
```

```
yosys> show
```

```
8. Generating Graphviz representation of design.
Writing dot description to `show.dot'.
Dumping module nor_gate to page 1.
```

```
yosys> exit
```

```
[OSS CAD Suite] D:\> \windows_10_cmake_Release_Graphviz-15.1.1-win64\Graphviz-15.1.1-win64\bin\dot.exe" -Tpng show.dot -o schematic.png
```

Generate & check and_gate schematic - Fig. 4(b)

